

Quadratic Equation Notes for IIT JEE

Quadratic Equation Notes for IIT JEE contain all the important formulas, identities, theorems, properties, and equations. Students are recommended to learn Quadratic Equation Notes for IIT JEE. Revising these Quadratic Equation Notes for IIT JEE will help students easily memorize the important formulas and main topics.

We provide complete notes of the Quadratic Equation for IIT JEE. Students can also easily download the Quadratic Equation Notes for IIT JEE pdf for free.

Quadratic Equation

A quadratic equation is a second-degree equation. The general form of the [quadratic equation](#) is $ax^2 + bx + c = 0$, where a , b , c are real numbers and $a \neq 0$. For example, $x^2 + 2x + 1 = 0$.

A polynomial is an algebraic expression containing many terms. A polynomial of degree two of the form $ax^2 + bx + c$ ($a \neq 0$) is called a quadratic expression in x . We can also say that a quadratic polynomial $f(x)$, when equated to zero, is called a quadratic equation.

Table of Contents:

- Roots of Quadratic Equation
- Sum and Product of Roots
- Nature of Roots
- Graph of Quadratic Equation
- Quadratic Expression in Two Variables
- Solved Examples
- Practice Problems
- Frequently Asked Questions

Roots of Quadratic Equation

The values of x which satisfy the quadratic equation are called the roots of the quadratic equation. Roots are also called zeros or solutions of a quadratic equation.

Solving Quadratic Equation

a) Using quadratic formula: If $ax^2 + bx + c = 0$ is the given quadratic equation, the roots are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

b) Factorisation method: Let $ax^2 + bx + c = a(x - \alpha)(x - \beta) = 0$, then $x = \alpha$ and $x = \beta$ will satisfy the given equation.

So factorizing the equation and equating each to zero gives the roots of the given equation.

Sum and Product of Roots

If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then
Sum of roots, $\alpha + \beta = -b/a$

Product of roots, $\alpha\beta = c/a$

Discriminant of Quadratic Equation

The discriminant of the quadratic equation $ax^2 + bx + c = 0$ is given by $D = b^2 - 4ac$. It is also denoted by Δ .

Nature of Roots

If $D = 0$, the roots are real and equal.

If $D > 0$, the roots are real and unequal.

If $D < 0$, the roots are imaginary and unequal.

If $D > 0$ and D is a perfect square, the roots are rational and unequal.

If $D > 0$ and D is not a perfect square, the roots are irrational and unequal.

Formation of Quadratic Equation with given roots

If α and β are the roots of the quadratic equation, then

$$(x - \alpha)(x - \beta) = 0$$

$$\Rightarrow x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\Rightarrow x^2 - (\text{sum of roots})x + \text{product of roots} = 0$$

Conjugate Roots: The irrational and complex roots of a quadratic equation with a rational coefficient always occur in pairs. So if one root is $\alpha + i\beta$, then the other root will be $\alpha - i\beta$.

Relation between Roots and Coefficients

If α and β are the roots of quadratic equation $ax^2 + bx + c = 0$, then

$$(a) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (b^2 - 2ac)/a^2$$

$$(b) \alpha - \beta = \sqrt{((\alpha + \beta)^2 - 4\alpha\beta)} = \pm\sqrt{(b^2 - 4ac)/a} = \pm\sqrt{D}/a$$

$$(c) \alpha^2 - \beta^2 = (\alpha + \beta)\sqrt{((\alpha + \beta)^2 - 4\alpha\beta)} = -b\sqrt{(b^2 - 4ac)}/a^2$$

$$(d) \alpha^3 - \beta^3 = (\alpha - \beta)^3 + 3\alpha\beta(\alpha - \beta) = (b^2 - ac)\sqrt{(b^2 - 4ac)}/a^3$$

$$(e) \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = -b(b^2 - 3ac)/a^3$$

$$(f) \alpha^2 + \alpha\beta + \beta^2 = (\alpha + \beta)^2 - \alpha\beta$$

Roots under special cases

Consider the quadratic equation $ax^2 + bx + c = 0$

(i) if $c = 0$, then one root is zero. Other root is $-b/a$.

(ii) If $b = 0$, the roots are equal but opposite in sign.

(iii) If $b = c = 0$, then both roots are zero.

(iv) If $a = c$, then the roots are reciprocal to each other.

(v) If $a + b + c = 0$, then one root is 1 and the second root is c/a .

(vi) If $a = b = c = 0$, then the equation will become an identity and will satisfy every value of x .

Nature of Factors of the Quadratic Expression

The nature of factors of the quadratic expression $ax^2 + bx + c$ is the same as the nature of roots of the corresponding quadratic equation $ax^2 + bx + c = 0$. Hence, the factors of the expression are

(i) Real and different, if $b^2 - 4ac > 0$.

(ii) Rational and different, if $b^2 - 4ac$ is a perfect square.

(iii) Real and equal, if $b^2 - 4ac = 0$.

(iv) Imaginary, if $b^2 - 4ac < 0$.

Condition for Common Roots

(a) Only one common root

Consider the following quadratic equations

$$a_1x^2 + b_1x + c_1 = 0$$

$$a_2x^2 + b_2x + c_2 = 0$$

Let α be the common root of the above quadratic equations.

So we can write

$$a_1\alpha^2 + b_1\alpha + c_1 = 0$$

$$a_2\alpha^2 + b_2\alpha + c_2 = 0$$

The condition for only one root common is

$$(c_1a_2 - c_2a_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)$$

(b) Both roots are common

The condition for both roots are common is

$$a_1/a_2 = b_1/b_2 = c_1/c_2$$

Graph of Quadratic Equation

Consider a quadratic equation $ax^2 + bx + c = 0$, where a , b , and c are real and $a \neq 0$.

The graph of a quadratic equation is a parabola.

If $a > 0$, then the graph of a quadratic equation will be concave upwards.

If $a < 0$, then the graph of a quadratic equation will be concave downwards.

The value of the discriminant determines whether the graph of a quadratic equation will:

- Intersect the x-axis at two points i.e. $b^2 - 4ac > 0$
- Just touches the x-axis i.e. $b^2 - 4ac = 0$
- Never intersects the x-axis i.e. $b^2 - 4ac < 0$

Case 1: If $a > 0$, $D > 0$

The graph of a quadratic equation will be concave upwards and will intersect the x-axis at two points α and β . The quadratic equation will have two real roots (α and β), and the curve will always lie above the x-axis.

Case 2: If $a > 0$, $D = 0$

The graph of a quadratic equation will be concave upwards and will intersect the x-axis at one point $(-b/2a)$. The quadratic equation will have two equal roots ($\alpha = \beta$).

Case 3: If $a > 0$, $D < 0$

The graph of a quadratic equation will be concave upwards and will not intersect the x-axis. The quadratic equation will have imaginary roots.

Case 4: If $a < 0$, $D > 0$

The parabola will be concave downwards and will intersect the x-axis at two points α and β . The quadratic equation will have two real roots.

Case 5: If $a < 0$, $D = 0$

The parabola will be concave downwards and will intersect the x-axis at one point. The quadratic equation will have two equal roots.

Case 6: If $a < 0$, $D < 0$

The parabola will be concave downwards and will not intersect the x-axis. The quadratic equation will have imaginary roots.

Maximum and Minimum Value

Consider the quadratic expression $ax^2 + bx + c$

a) If $a < 0$, then the expression has the greatest value at $x = -b/2a$. The maximum value is $-D/4a$.

b) If $a > 0$, then the expression has the least value at $x = -b/2a$. The minimum value is $-D/4a$.

Quadratic Expression in Two Variables

The general form of a quadratic equation in two variables x and y is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c.$$

To resolve the above expression into 2 linear rational factors, the condition is

$$\Delta = 0$$

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \text{ and } h^2 - ab > 0$$

This is called the discriminant of the given expression.

Important Points to Remember

(i) If α is a root of the equation $f(x) = 0$, then the polynomial $f(x)$ is exactly divisible by $(x - \alpha)$. This implies that $(x - \alpha)$ is a factor of $f(x)$.

(ii) Every equation of n th ($n \geq 1$) degree has exactly n roots, and if the equation has more than n roots, then it is an identity.

(iii) If roots of quadratic equation $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ are in the same ratio, (i.e., $\alpha_1/\beta_1 = \alpha_2/\beta_2$) then $b_1^2/b_2^2 = a_1c_1/a_2c_2$.

(iv) Quadratic equations having a modulus sign are solved considering both positive and negative values of the quantity containing the modulus sign. The roots of the given equation are those values among the values of the variable thus obtained which satisfy the given equation.

(v) If one root is k times the other root of a quadratic equation $ax^2 + bx + c = 0$, then $(k + 1)^2/k = b^2/ac$.

Solved Examples

Example 1.

If α and β are the roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{2009} + \beta^{2009} =$

(a) -2

- (b) -1
- (c) 1
- (d) 2

Solution:

Given $x^2 - x + 1 = 0$

Using the quadratic formula, we get $x = (1 \pm i\sqrt{3})/2$

$\alpha = -\omega, \beta = -\omega^2$

$\alpha^{2009} + \beta^{2009} = -\omega^{2009} + (-\omega^2)^{2009}$

$= -(\omega^2 + \omega)$

$= 1$

Hence option c is the answer.

Example 2:

If p and q are the roots of the equation $x^2 + px + q = 0$, then

- (a) $p = 1, q = -2$
- (b) $p = 0, q = 1$
- (c) $p = -2, q = 0$
- (d) $p = -2, q = 1$

Solution:

Given $x^2 + px + q = 0$

Sum of roots, $p + q = -p$

Product of roots, $pq = q$

$\Rightarrow p = 1$

$1 + q = -1$

$\Rightarrow q = -2$

So $p = 1, q = -2$.

Hence option a is the answer.

Example 3:

Let two numbers have arithmetic mean 9 and geometric mean 4. Then these numbers are the roots of the quadratic equation

- (a) $x^2 + 18x + 16 = 0$
- (b) $x^2 - 18x - 16 = 0$
- (c) $x^2 + 18x - 16 = 0$
- (d) $x^2 - 18x + 16 = 0$

Solution:

Let m and n be the numbers.

$(m + n)/2 = 9$

$(m + n) = 18$

$\sqrt{mn} = 4$

$mn = 16$

So the equation is $x^2 - (\text{sum of roots})x + \text{product} = 0$

$\Rightarrow x^2 - 18x + 16 = 0$

Hence option d is the answer.

Example 4:

If one of the roots of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of q is

- (a) $49/4$
- (b) 4
- (c) 3
- (d) 12

Solution:

Given $x^2 + px + 12 = 0 \dots(i)$

Since 4 is a root of (i)

$$4^2 + 4p + 12 = 0$$

$$\Rightarrow 4p = -28$$

$$\Rightarrow p = -7$$

Given $x^2 + px + q = 0$ has equal roots.

So $D = 0$

$$p^2 - 4q = 0$$

$$\Rightarrow 49 - 4q = 0$$

$$\Rightarrow q = 49/4$$

Hence option a is the answer.

Example 5:

The value of “a” for which the sum of the squares of the roots of the equation $x^2 - (a - 2)x - a - 1 = 0$ assumes the least value is

- (a) 1
- (b) 0
- (c) 3
- (d) 2

Solution:

Given $x^2 - (a - 2)x - a - 1 = 0$

Sum of roots, $\alpha + \beta = (a - 2)$

Product of roots, $\alpha\beta = -a - 1$

$$\text{Sum of squares of roots} = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= a^2 - 2a + 6$$

$$= (a - 1)^2 + 5$$

$$\Rightarrow a = 1$$

Hence option a is the answer.

Practice Problems

1. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has

- (a) No real roots
- (b) Exactly one root
- (c) Infinite number of real roots
- (d) Exactly four real roots

2. If α and β are the distinct roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{101} + \beta^{107}$ is equal to

- (a) 0
- (b) 1
- (c) -1
- (d) 2

3. If the roots of the equation $bx^2 + cx + a = 0$ be imaginary, then for all real values of x , the expression $3b^2x^2 + 6bcx + 2c^2$ is

- (a) greater than $4ab$
- (b) less than $4ab$
- (c) greater than $-4ab$
- (d) less than $-4ab$

4. If $x^2 + px - 444p = 0$ has integral roots, where p is a prime number, then the value of p is

- (a) 2
- (b) 2, 3 and 37
- (c) 3
- (d) 37

5. If the roots of the equation $x^2 - bx + c = 0$ are two consecutive integers, then $b^2 - 4ac$ equals

- (a) -2
- (b) 3
- (c) 2
- (d) 1

Frequently Asked Questions

What are the applications of quadratic equations in daily life?

Quadratic equations find many applications in daily life. We use the quadratic formula for calculating area, determining a product's profit, and calculating the speed of an object.

Write the general form of a quadratic equation.

The general form of a quadratic equation is $ax^2 + bx + c = 0$, where a , b , c are real numbers and $a \neq 0$.

What do you mean by the roots of a quadratic equation?

The values of x which satisfy the quadratic equation are called the roots of the quadratic equation.

What are the 4 methods to solve quadratic equations?

The four methods of solving a quadratic equation are factoring, using the square roots, completing the square method, and the quadratic formula.

What is the graph of a quadratic function?

The graph of a quadratic function is a U-shaped curve, also called a parabola.